

Medium Article RAG Assistant

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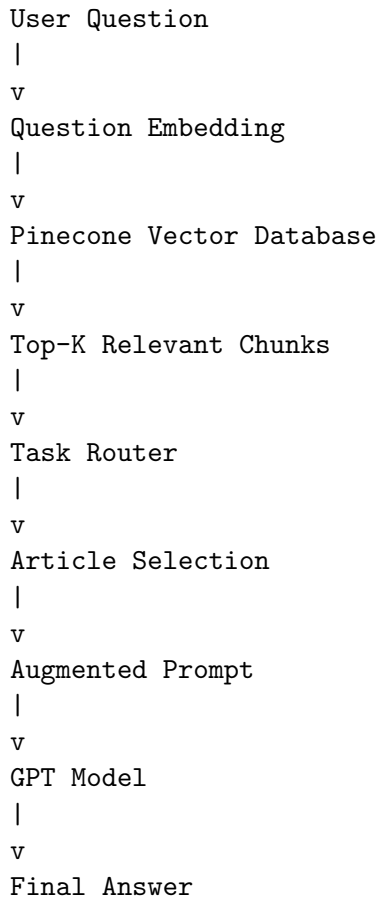
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1 Introduction

The goal of this project is to develop a Retrieval-Augmented Generation (RAG) assistant specialized in Medium articles. The assistant answers user questions using only information retrieved from the provided dataset, without relying on external knowledge.

2 System Architecture

The system follows an enhanced Retrieval-Augmented Generation (RAG) pipeline.



In addition to standard RAG retrieval, the system includes a Task Router and an Article Selection layer. These components help distinguish between article-focused tasks and general question-answering tasks, reducing irrelevant context and improving answer quality.

3 Models Used

The system relies on two different models provided through the course platform:

- **4UHRUIN-text-embedding-3-small** – used to convert article chunks and user queries into dense vector representations for semantic retrieval.
- **4UHRUIN-gpt-5-mini** – used as the generation model. Retrieved context is combined with the user question to generate the final answer.

This separation follows the standard RAG architecture, where one model is responsible for retrieval and another is responsible for answer generation.

4 Dataset Exploration and Hyperparameter Selection

Before building the RAG pipeline, we analyzed the provided Medium articles dataset. The dataset contains 7,682 articles, with no missing values in the relevant fields.

The token analysis showed that the average article length is approximately 1,367 tokens, with a median length of 1,105 tokens. Based on the course recommendations for long-form articles, we selected a chunk size of 800 tokens and an overlap ratio of 20

The selected hyperparameters are:

- **Chunk size:** 800 tokens
- **Overlap ratio:** 0.2
- **Initial retrieval top-k:** 8

5 Chunking Strategy

The dataset articles were split using a token-based chunking strategy. Token-based splitting was preferred over character-based splitting to ensure consistency with the embedding model's tokenization scheme.

After indexing the complete dataset, the final knowledge base contained 18,661 embedded chunks generated from 7,682 Medium articles.

6 Task-Aware Retrieval

During development we observed that chunk-level retrieval sometimes returned passages from multiple unrelated articles.

To address this issue, we implemented a task-aware retrieval mechanism. The system first classifies the query into one of the following categories:

- Fact Retrieval
- Article Listing
- Article Summarization
- Article Recommendation
- General Question Answering

For article-focused tasks, retrieved chunks are grouped by article and ranked at the article level before being sent to the language model. For general question-answering tasks, information from multiple articles may be combined when appropriate.

7 Final Knowledge Base

The complete Medium dataset was indexed into Pinecone.

- Articles indexed: 7,682
- Generated chunks: 18,661
- Embedding model: 4UHRUIN-text-embedding-3-small
- Vector database: Pinecone
- Similarity metric: Cosine Similarity

8 Conclusion

A complete RAG assistant for Medium articles was developed using OpenAI embeddings, Pinecone vector search, and GPT-based answer generation. The final system combines semantic retrieval with task-aware article selection, enabling accurate retrieval, summarization, recommendation, and question-answering over the entire Medium corpus.